Notes: Jung, Engle, & Berner (JFE, 2025)
CRISK: Measuring the Climate Risk Exposure of the Financial System

Contents

1 Big picture

- **Question.** Are large financial institutions sufficiently capitalized to absorb losses under severe climate transition stress?
- **Main object.** The paper introduces CRISK, the expected capital shortfall of a financial firm conditional on a climate transition stress event.
- **Incremental object.** Marginal CRISK, or mCRISK, isolates the additional capital shortfall attributable specifically to climate stress.
- **Climate beta.** The key input is each bank's stock-return sensitivity to a climate transition risk factor.
- **Validation.** Climate betas are validated against granular Y-14 loan portfolio data.
- **Credit channel.** High-climate-beta borrowers have higher loan probabilities of default, and banks adjust brown loan prices and quantities only slowly.
- **Systemic result.** Aggregated CRISK and mCRISK show that transition risk can generate large capital-shortfall estimates, concentrated especially in the US banking sector.
- **Economic takeaway.** Climate transition risk is both a market-risk problem and a credit-risk problem.

2 Data and measurement

2.1 Climate transition risk factors

The paper constructs several climate transition risk factors: stranded asset, emission, brown-minus-green (BMG), and climate-efficient portfolio (CEP). Each factor is designed so that its value declines when transition risk rises.

Interpretation. The factors convert abstract transition-risk news into traded return series that can be used in betas and stress scenarios.

2.2 Climate beta of banks

Climate beta is the sensitivity of a bank's stock return to the climate transition risk factor after controlling for market returns. The paper estimates it dynamically because banks' balance sheets and market perceptions change over time.

Interpretation. A climate beta is a market-implied measure of how much investors think a bank is exposed to transition risk.

2.3 Loan portfolio validation data

Using Y-14 data, the paper constructs loan portfolio climate beta as a weighted average of borrower or industry climate betas.

$$\text{Loan Portfolio Climate Beta}_{it} = \sum_j w_{ij,t} \beta_{jt}^{Climate}.$$

Interpretation. This balance-sheet measure tests whether market-based climate beta reflects actual lending exposure to brown industries.

2.4 CRISK, mCRISK, and S&CRISK

CRISK is expected capital shortfall conditional on climate stress. mCRISK subtracts the capital shortfall under no climate stress. S&CRISK combines climate stress with broad market stress.

Interpretation. These measures separate total capital shortfall, marginal climate exposure, and compound climate-market stress.

3 Models and empirical specifications

3.1 Climate factor event-study validation

$$CF_t = \alpha + \sum_{n=0}^{5} \gamma_n shock_{t-n} + bMKT_t + \varepsilon_t.$$

Interpretation. The coefficient path tests whether the climate factors move around climate-change events in the expected direction.

3.2 Climate beta estimation

$$r_{it} = \beta_{it}^{MKT} MKT_t + \beta_{it}^{Climate} CF_t + \varepsilon_{it}.$$

Interpretation. The climate beta is the bank stock return sensitivity to the climate factor after controlling for the market.

3.3 Validation regression

$$\beta_{it}^{Climate} = \alpha + b\,\text{LoanPortfolioClimateBeta}_{it} + Controls_{it} + \delta_i + \gamma_t + \varepsilon_{it}.$$

Interpretation. A positive coefficient links market-implied bank climate exposure to actual loan portfolio composition.

3.4 CRISK formulas

$$CS_{it} = k(D_{it} + W_{it}) - W_{it}, \qquad CRISK_{it} = E[CS_{i,t+h} \mid \text{climate stress}],$$
$$CRISK_{it} = kD_{it} - (1-k)W_{it}(1 - LRMES_{it}), \qquad mCRISK_{it} = (1-k)W_{it}LRMES_{it}.$$

Interpretation. CRISK is high when a firm is large, levered, and expected to lose substantial equity value under climate stress.

3.5 Compound stress

$$S\&CRISK_{it} = kD_{it} - (1-k)W_{it} \exp\left(\beta_{it}^{Climate} \log(1-\theta^{Climate}) + \beta_{it}^{MKT} \log(1-\theta^{MKT})\right).$$

Interpretation. S&CRISK captures joint climate and broad market stress.

4 Identification and empirical design

- **Factor validation.** Climate factors are tested around climate-change events.
- **Balance-sheet validation.** Y-14 loan portfolios test whether market-based bank climate beta reflects lending exposure to brown industries.
- **Credit-risk channel.** Loan PDs rise with lagged climate beta.
- **Slow adjustment.** Loan spread and quantity adjustments occur slowly after brown-industry climate betas rise.
- **Out-of-sample check.** Pre-COVID climate betas predict COVID drawdowns and mCRISK.

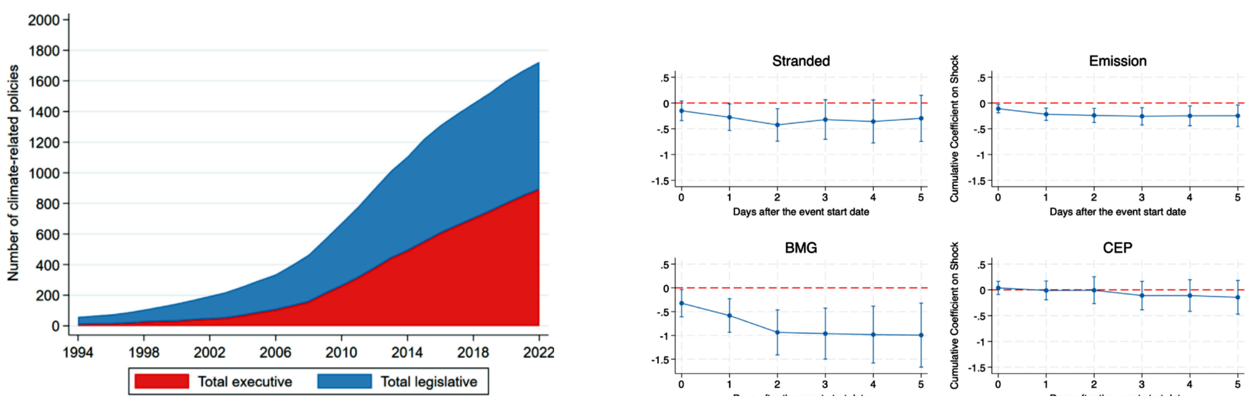
5 Tables and figures with expanded interpretation

5.1 Fig. 1–2: policy growth and factor validation

Read: Fig. 1 shows rapid growth in climate-related policies. Fig. 2 shows that the stranded, emission, and BMG climate factors move negatively around greener transition events.

Interpretation. This block validates the front end of the CRISK framework: the factors are related to climate-transition news.

Economic message. Without factor validity, climate betas and CRISK would be difficult to interpret.



(a) Global climate policies

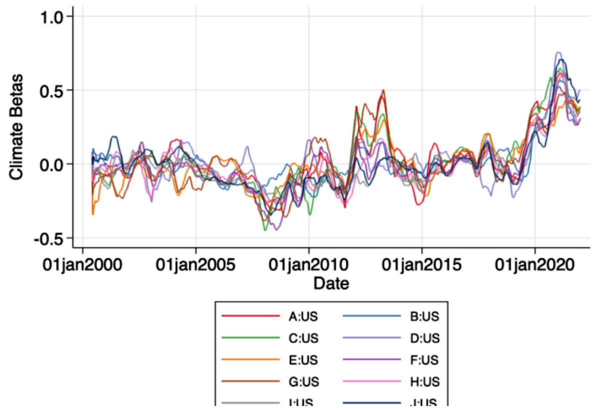
(b) Climate factor event responses

5.2 Fig. 3–4 and Table 1: climate beta and loan portfolio validation

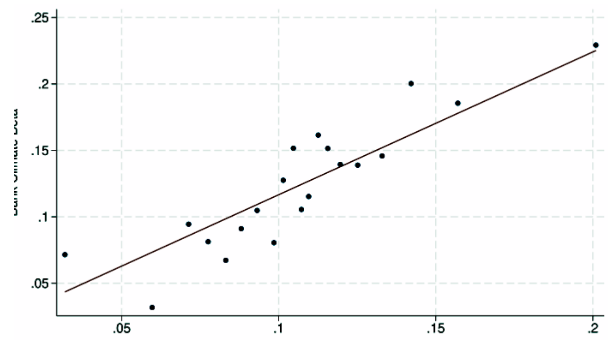
Read: Bank climate betas rise around 2019–2020. Bank climate beta is strongly aligned with loan portfolio climate beta.

Interpretation. This is the most important validation block because it connects market-implied exposure to actual loan portfolio composition.

Economic message. Banks with loan books tilted toward transition-exposed industries have higher market-implied transition risk.



(a) Top US bank climate betas



(b) Bank beta vs. loan portfolio beta

Table 1
Bank climate beta and loan portfolio climate beta Quarterly data from 2012:Q2 to 2021:Q4 for listed US banks in Y-14. Standard errors are clustered by banks.

	(1) Climate beta	(2) Climate beta	(3) Climate beta	(4) Climate beta
Loan Portfolio Climate Beta	1.718***	1.524***	1.400***	1.112***
	(5.67)	(5.80)	(6.37)	(3.58)
Log Assets		0.0109	0.389***	0.0827
		(0.67)	(6.25)	(1.14)
Leverage		3.958***	1.395	−0.810
		(3.24)	(1.19)	(−0.75)
ROA		8.595***	5.947***	2.345*
		(4.84)	(4.56)	(2.10)
Loans/Assets		0.115	−0.222	−0.286
		(0.94)	(−0.53)	(−1.06)
Deposits/Assets		0.232**	0.0422	−0.550**
		(2.37)	(0.18)	(−2.42)
Loan Loss Reserves/Loans		−2.155	5.367***	3.068*
		(−0.88)	(4.48)	(1.74)
Non-interest Income/Net Income		0.00144	0.00184	0.00157
		(0.90)	(1.39)	(1.11)
Market Beta		0.139***	0.113***	0.00793
		(4.62)	(5.90)	(0.41)
Book/Market		0.202***	0.113***	−0.00816
		(4.19)	(4.19)	(−0.21)
N	666	666	666	666
Bank Controls	N	Y	Y	Y
Bank FE	N	N	Y	Y
Year FE	N	N	N	Y
Adj R2	0.314	0.429	0.592	0.701

t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

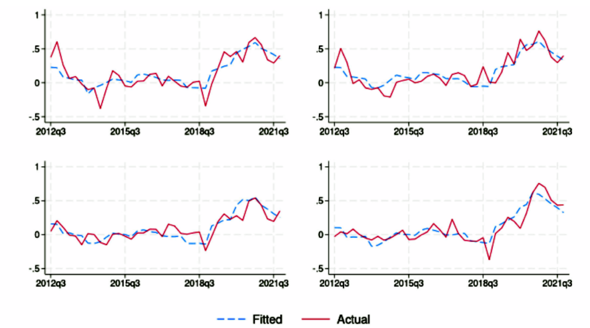
Figure 3: Bank climate beta and loan portfolio climate beta

5.3 Fig. 5–7: fitted bank betas, brown industries, and default risk

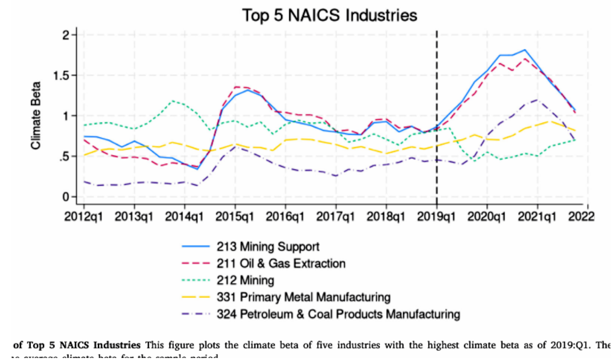
Read: The fitted climate betas track actual betas, brown industries have high climate betas, and loan PD rises with lagged climate beta.

Interpretation. These visuals strengthen the credit-channel mechanism linking transition-exposed borrowers to bank risk.

Economic message. Transition risk enters banks through loan portfolios and borrower default risk.



(a) Fitted vs. actual bank beta



(b) Top brown-industry climate betas

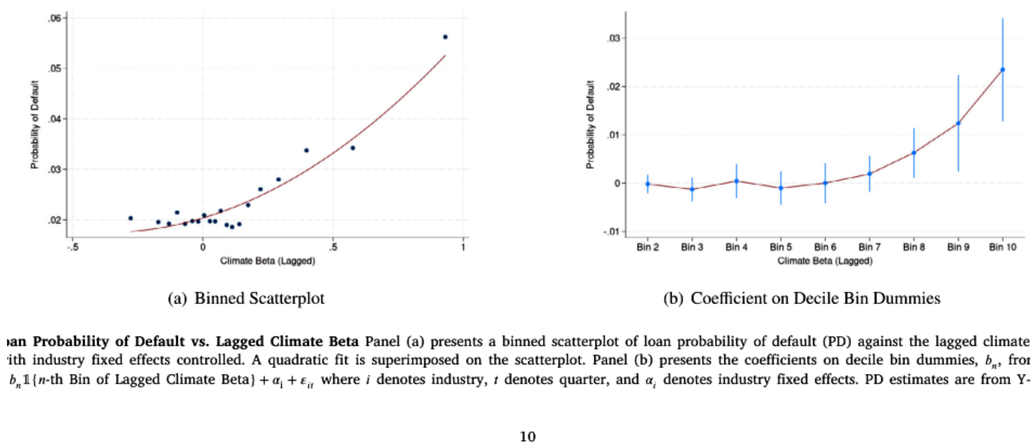


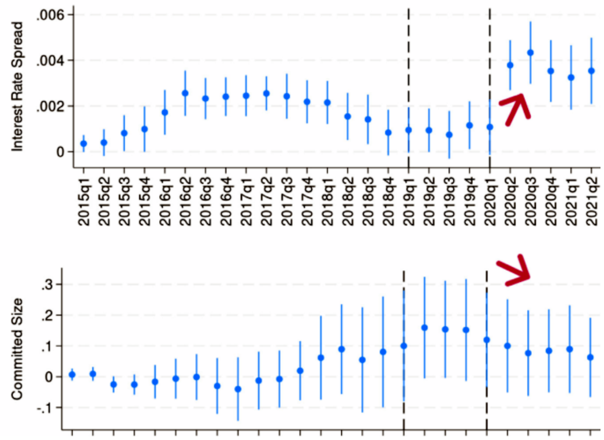
Figure 5: Loan probability of default vs. climate beta

5.4 Fig. 8–9: loan adjustment and climate-policy environment

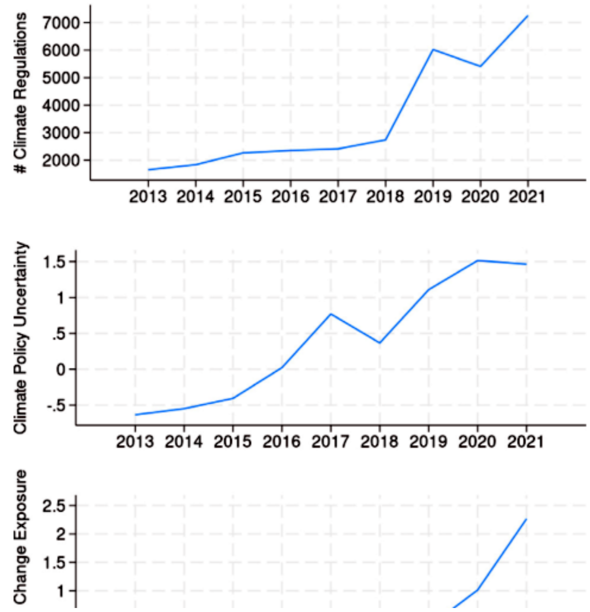
Read: Brown loan spreads rise and committed quantities fall after the rise in brown-industry climate betas. Climate regulation and policy uncertainty rise around 2019–2021.

Interpretation. Banks appear to adjust loan pricing and quantities, but only slowly.

Economic message. Slow adjustment can leave banks exposed during disorderly transition shocks.



(a) Brown loan price and quantity



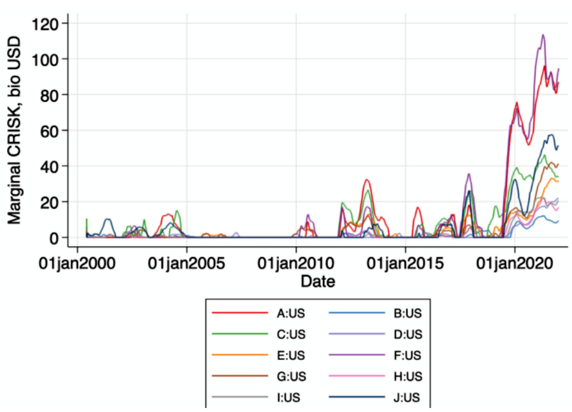
(b) Climate regulation and policy indices

5.5 Fig. 10–11 and Table 2: bank-level CRISK and decomposition

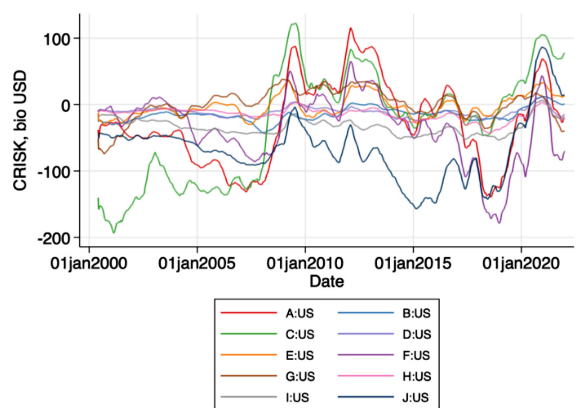
Read: mCRISK and CRISK rise sharply in 2020. Table 2 decomposes the 2020 increase into debt, equity, and risk components.

Interpretation. mCRISK isolates incremental climate stress, while CRISK measures total expected shortfall.

Economic message. Climate transition stress can create economically large capital-shortfall estimates for major banks.



(a) Marginal CRISK



(b) CRISK

CRISK decomposition (US Banks) CRISK(t) is the bank's CRISK at the end of 2020, and CRISK($t-1$) is CRISK at the end of year 2019. dCRISK= CRISK(t)-CRISK($t-1$) is the change in CRISK during 2020. dDEBT is the contribution of the firm's debt to CRISK. dEQUITY is the contribution of the firm's equity position on CRISK. dRISK is the contribution of increase in volatility or correlation to CRISK. All amounts are in billions USD. Top 4 banks include F:US, A:US, C:US, and J:US.

Bank	CRISK(t-1)	CRISK(t)	dCRISK	dDEBT	dEQUITY	dRISK
F:US	−146.58	−9.49	137.08	37.63	35.65	63.80
A:US	−52.35	43.80	96.15	24.63	35.37	36.15
C:US	13.34	93.70	80.36	17.49	29.85	33.03
J:US	−47.11	70.18	117.30	−0.84	70.56	47.57
E:US	9.86	22.56	12.70	9.90	−6.72	9.52
G:US	4.09	−6.68	−10.77	3.65	−27.91	13.50
I:US	−41.33	−5.32	36.01	4.13	16.21	15.67
H:US	−26.66	−9.60	17.06	3.80	4.97	8.29
B:US	−7.42	7.40	14.82	4.11	5.94	4.77
D:US	−10.48	1.53	12.00	3.25	0.22	8.54
Top 4	.	.	430.89	78.91	171.43	180.55

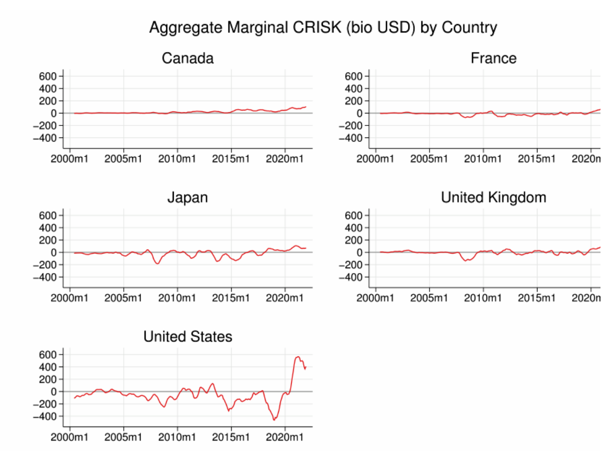
Figure 8: CRISK decomposition

5.6 Fig. 12–15: aggregate systemic risk

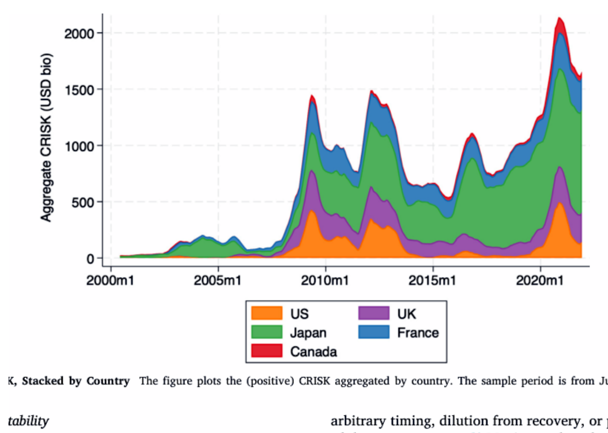
Read: Aggregate climate risk is largest for the US and concentrated in banks rather than broker-dealers or insurers.

Interpretation. The aggregation shows concentration across countries and financial sectors.

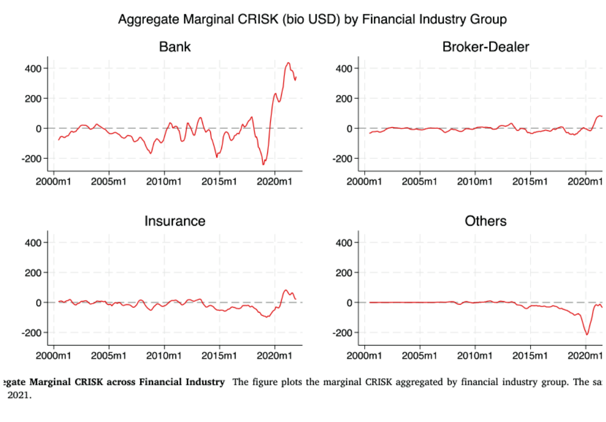
Economic message. Systemic climate transition risk is not evenly distributed; US banks are central in the paper's main scenario.



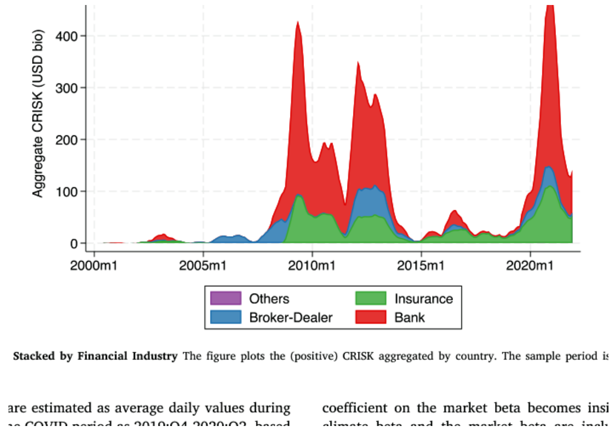
(a) mCRISK by country



(b) CRISK by country



(a) mCRISK by industry



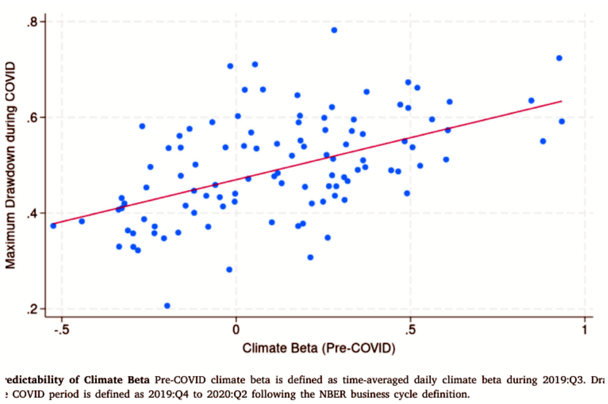
(b) CRISK by industry

5.7 Fig. 16–17 and Tables 3–4: out-of-sample predictability

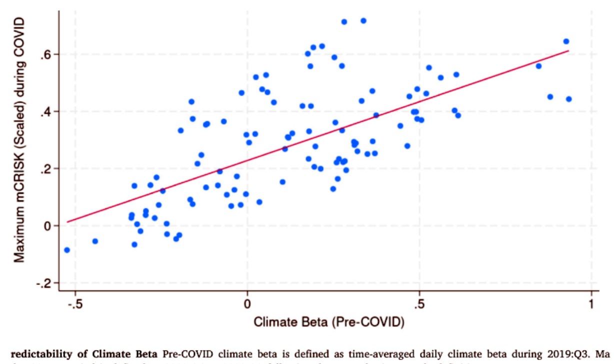
Read: Pre-COVID climate beta predicts COVID-period drawdowns and maximum mCRISK. Tables 3 and 4 confirm that climate beta remains significant after controlling for market beta.

Interpretation. These are out-of-sample validation tests.

Economic message. Climate beta contains stress-vulnerability information that market beta alone does not capture.



(a) Drawdown predictability



(b) mCRISK predictability

Cross-sectional predictability of climate beta Pre-COVID beta is defined as time-averaged daily beta during 2019:Q3. Drawdown, denoted as DD, is defined as the $|trough - peak|/peak$, and the COVID period is defined as 2019:Q4 to 2020:Q2 following the NBER business cycle definition.

	(1) Max DD (COVID)	(2) Max DD (COVID)	(3) Max DD (COVID)
Climate Beta (Pre-COVID)	0.180^{***}		0.176^{***}
	(6.18)		(5.13)
Market Beta (Pre-COVID)		0.111^{***}	0.00880
		(3.05)	(0.23)
Constant	0.474^{***}	0.381^{***}	0.465^{***}
	(48.74)	(9.79)	(12.07)
N	105	105	105
Adj R2	0.264	0.0741	0.257

t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

(a) Drawdown regression

Table 4
Cross-sectional predictability of climate beta Pre-COVID beta is defined as time-averaged daily beta during 2019:Q3. Marginal CRISK (mCRISK) is scaled by the market cap. The COVID period is defined as 2019:Q4 to 2020:Q2 following the NBER business cycle definition.

	(1) Max mCRISK (COVID)	(2) Max mCRISK (COVID)	(3) Max mCRISK (COVID)
Climate Beta (Pre-COVID)	0.414^{***}		0.450^{***}
	(9.26)		(8.62)
Market Beta (Pre-COVID)		0.184^{***}	−0.0772
		(2.84)	(−1.33)
Constant	0.235^{***}	0.0962	0.311^{***}
	(15.76)	(1.38)	(5.29)
N	105	105	105
Adj R2	0.449	0.0636	0.453

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

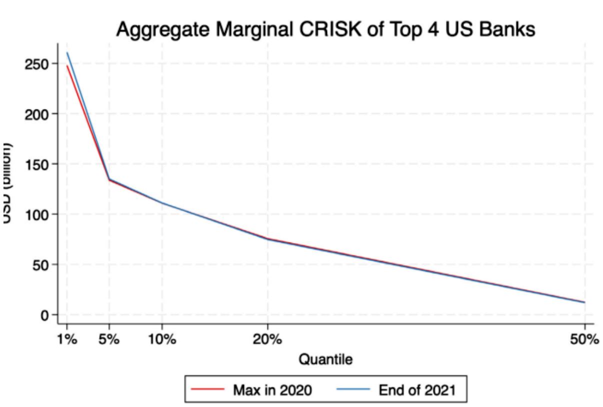
(b) mCRISK regression

5.8 Fig. 18–20: scenario severity and compound stress

Read: More severe climate stress scenarios imply much larger mCRISK; combining climate and market stress produces larger S&CRISK exposures.

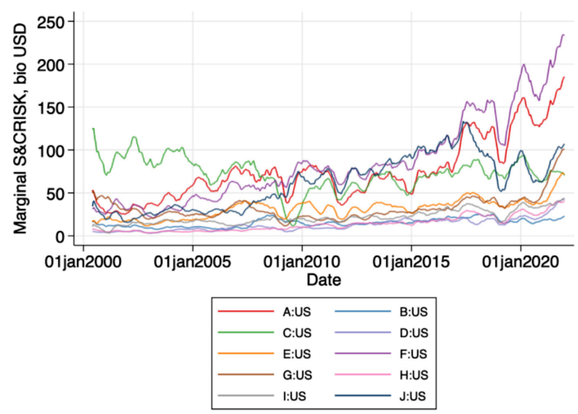
Interpretation. Magnitudes are scenario-dependent and depend on whether climate stress coincides with broad market stress.

Economic message. A disorderly transition is most dangerous when it coincides with market-wide stress.



igure plots the aggregate marginal CRISK of the top 4 US banks across different severities of the scenario.

(a) Scenario severity sensitivity



(of US Banks The sample banks are the top 10 large US banks by the average total assets in 2019. The sample peri

(b) Marginal S&CRISK

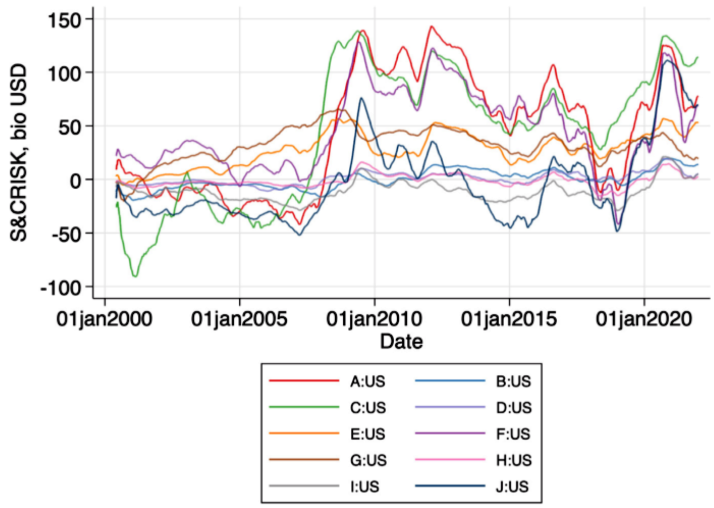


Figure 14: S&CRISK of US banks

6 Short summary

- The paper develops CRISK, a market-based expected capital-shortfall measure for climate transition stress.
- It constructs climate transition factors and validates them through event studies.
- Bank climate beta measures stock-return sensitivity to transition risk factors.
- Loan portfolio climate beta strongly predicts market-based climate beta.
- Loan default probabilities rise with lagged borrower climate beta.
- mCRISK and CRISK rise sharply for major US banks around 2020 in the stranded-asset scenario.
- Pre-COVID climate betas predict COVID drawdowns and maximum mCRISK.
- Scenario analysis shows that joint climate-market stress can produce larger exposures than climate stress alone.

7 Expanded conclusion

7.1 Core conclusion

The paper turns climate transition risk into a measurable financial-stability object: expected capital shortfall.

7.2 Evidence chain

The evidence chain proceeds from factor validation to market beta estimation, loan portfolio validation, credit-risk channel tests, CRISK applications, aggregate systemic measures, and out-of-sample predictability.

7.3 Economic mechanism

Climate policy and transition expectations reprice brown industries. Borrowers in those industries become riskier. Banks with larger exposures have higher climate betas and can face higher expected capital shortfalls.

7.4 What not to overinterpret

CRISK is a stress-test measure, not a forecast of a realized event. Magnitudes depend on factor choice, stress severity, horizon, and capital ratio assumptions. Climate beta is market-based and reflects investor beliefs as well as real exposures.

7.5 Broader implications

The framework is useful for supervisors because it combines market prices with balance-sheet validation. It can be updated more frequently than disclosure-based measures and can be aggregated across firms and sectors.

7.6 Final takeaway

Climate transition risk can be systemic when bank balance sheets are exposed to borrowers whose values fall in a disorderly transition. CRISK measures how much capital may be needed if that risk materializes.